

Robotics-for-Cacao Barter

Follow-up action item · TrueSight DAO / Agroverse · 2026-08-06

Context. Three field priorities: harvesting cacao, cutting off witches' broom (*vassoura-de-bruxa*, *Moniliophthora perniciosa*), and cacao roasting flavor-profile control. A robotics design expert partner is on board. Working-prototype reference: **Frasky**, the IIT vineyard robot (Reuters video, <https://youtu.be/hg8qYrjyYCU>). This follow-up turns that introduction into a concrete pilot.

1 · The problem on the farm

- **Witches' broom** is the single highest-leverage canopy disease in Bahia cacao: infected brooms must be pruned out **and removed from the plantation** in dry season, or spores reinfect — cutting yields by 30–50% on affected trees.
- **Labor shortage:** pruning and selective pod harvest are manual, seasonal, and hard to staff.
- **Cash constraint:** smallholders have little cash but hold **cacao** — the natural payment asset.

2 · Working prototype reference — Frasky (IIT)

Video (Reuters): <https://youtu.be/hg8qYrjyYCU> — "A hands-on robot tends grapes in Italy's vineyards"

- Built by the **Italian Institute of Technology + Bergamo-region agricultural stakeholders** (roboticist Dr. Manuel Catalano).
- Autonomously **monitors grape clusters** (camera + digital map of every cluster's location), **manipulates plants** (robotic arm), and **applies targeted spray treatments**.
- Designed for precision agriculture, **labor-shortage relief**, and sustainability — field-tested in vineyards (Reuters, Nov 2025).

Proof point: a permanent tree/vine crop's canopy work can be robotized. Cacao is the same category — this is the working prototype we point partners and farmers to.

Frasky capability (grape)	Cacao adaptation	Feasibility
Cluster monitoring via camera	Broom / pod detection via vision	■ tractable
Manipulate plants (robotic arm)	Broom pruning with cutter arm	■ tractable
Targeted spray treatments	Targeted biocontrol on brooms	■ tractable
—	Selective pod <i>harvest</i> (varied heights: trunk + branches)	■■■ harder R&D

3 · Second application — scalable roast control via particle detection

The gap. Cacao roasting — the step that defines the flavor profile — is still largely a **skilled-craft process**: roast curves are tuned by experienced tasters, batch by batch. That does not scale beyond artisan volumes,

and it is hard to reproduce consistently across batches, roasters, and operators.

The opportunity. Machine vision + spectroscopy can detect bean/particle state in real time and close the loop on the roast:

Detection approach	What it measures	Status
Machine vision (camera)	Bean color, surface browning, particle size distribution	■ proven in food lines
Hyperspectral / SWIR imaging	Moisture, roast degree, bean defects (slaty, moldy, under/over-fermented)	■ proven, conveyor-scalable
Portable NIR spectroscopy	Chemical markers of roast degree, in-line	■ proven, low-cost
Electronic nose + ANN	Volatile profile → predicted roast degree (94.4% accuracy reported)	■ proven

Why now / why us. These methods exist but are sold as expensive lab equipment. A robotics + ML partner can build a **low-cost, in-line particle/roast detector** tuned to the beans and roast profiles the Agroverse network actually runs — turning roast flavor control from craft into a **scalable, measurable process**.

This is the **second application** for the same robotics partner: field robots (pruning/harvest) + processing sensor systems (roast control).

4 · The proposal (barter model)

Offer robotics services to Bahia/Pará farmers **in exchange for cacao as payment**:

Role	Who
Robot design & build	Robotics design expert (Gianluca)
Farmer network & deployment	Agroverse — Bahia operations (Ilhéus)
Market for cacao received	TrueSight DAO / Agroverse (US import live)

Honest MVP scoping. Broom *detection + pruning* (or monitoring + targeted treatment, à la Frasky) is tractable now. *Selective cacao pod harvesting* is harder R&D — set expectations so the pilot stays deliverable. Roast-particle detection is a parallel, lower-field-risk track.

5 · Pilot plan (1 season, measured)

1. **Scoping call with the robotics partner** — share the **Frasky video** (<https://youtu.be/hg8qYrjyYCU>) as the reference prototype; define MVP build, season timing, and cost basis; evaluate the roast-detector track.
2. **Farmer pitch (PT-BR)** — value prop: labor shortage, broom yield loss, **no cash outlay**, paid in cacao at fair market BRL.
3. **2–3 test farms** in the Bahia (Ilhéus) network.

4. **Barter accounting** — value cacao received at market BRL/USD; record as INVENTORY MOVEMENT + purchase into the Main Ledger (see SUPPLY_CHAIN_AND_FREIGHTING.md).

5. **Measure, then scale** — % broom removed, harvest kg, labor-hours saved; roast batch-to-batch consistency. No scaling on unmeasured evidence.

6 · Next steps (owners)

- **Approve outreach copy** and introduce the robotics partner (WhatsApp).
- **Draft scoping-call question list** (incl. Frasky feasibility + roast-detector questions) and the PT-BR farmer pitch.
- **Robotics partner** — feasibility call; MVP definition.
- **Agroverse Brazil operations** — identify 2–3 test farms; coordinate season logistics.

7 · Tie-back to mission

Every broom pruned, every robot-assisted harvest, and every batch of cacao roasted to a reproducible flavor profile keeps more trees producing and more **Amazon rainforest standing** — the DAO's 10,000-hectare regeneration goal is served by making cacao farming viable without clearing new land.

Filed in agentic_ai_context/OPEN_FOLLOWUPS.md (PR #733) · nothing sent to any party yet — awaiting governor approval of outreach copy.